SG No. 225 O_{h}^{5} $Fm\bar{3}m$ [cubic]

* plus set: $+[0,\,0,\,0]$, $+\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$, $+\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$, $+\left[\frac{1}{2},\,\frac{1}{2},\,0\right]$

Table 1: Wyckoff site: `4a`, site symmetry: `m-3m`

No.	position	mapping
1	[0, 0, 0]	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,`
		`25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]`

Table 2: Wyckoff site: `4b`, site symmetry: `m-3m`

No.	position	mapping
1	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,`
		`25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]`

Table 3: Wyckoff site: `8c`, site symmetry: `-43m`

No.	position	mapping
1	$\left[\frac{1}{4},\,\frac{1}{4},\,\frac{1}{4}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,37,38,39,40,41,42,43,44,45,46,47,48]`
2	$\left[\frac{1}{4},\,\frac{1}{4},\,\frac{3}{4}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36]`

Table 4: Wyckoff site: `24d`, site symmetry: `m.mm`

No.	position	mapping
1	$\left[0,\,\frac{1}{4},\,\frac{1}{4}\right]$	`[1,4,18,19,25,28,42,43]`
2	$\left[0,\,\frac{3}{4},\,\frac{1}{4}\right]$	`[2,3,17,20,26,27,41,44]`
3	$\left[\frac{1}{4},\,0,\,\frac{1}{4}\right]$	`[5,8,14,15,29,32,38,39]`
4	$\left[\frac{1}{4},\,0,\,\frac{3}{4}\right]$	`[6,7,13,16,30,31,37,40]`
5	$\left[\frac{1}{4},\,\frac{1}{4},\,0\right]$	`[9,12,21,24,33,36,45,48]`
6	$\left[\frac{3}{4},\,\frac{1}{4},\,0\right]$	`[10,11,22,23,34,35,46,47]`

Table 5: Wyckoff site: `24e`, site symmetry: `4m.m`

No.	position	mapping
1	$[x,\,0,\,0]$	`[1,4,17,20,26,27,42,43]`
2	$[-x,\,0,\,0]$	`[2,3,18,19,25,28,41,44]`
3	$[0,\,x,\,0]$	`[5,8,13,16,30,31,38,39]`
4	$[0,\,-x,\,0]$	`[6,7,14,15,29,32,37,40]`
5	$[0,\,0,\,x]$	`[9,12,22,23,34,35,45,48]`
6	$[0,\,0,\,-x]$	`[10,11,21,24,33,36,46,47]`

Table 6: Wyckoff site: `32f`, site symmetry: `.3m`

No.	position	mapping
1	$[x,\ x,\ x]$	`[1,5,9,38,43,48]`
2	$[-x,\ -x,\ x]$	`[2,7,12,37,41,45]`
3	$[-x,\ x,\ -x]$	`[3,8,10,39,44,46]`
4	$[x,\ -x,\ -x]$	`[4,6,11,40,42,47]`
5	$[x,\ x,\ -x]$	`[13,17,21,26,31,36]`
6	$[-x,\ -x,\ -x]$	`[14,19,24,25,29,33]`
7	$[x,\ -x,\ x]$	`[15,20,22,27,32,34]`
8	$[-x,\ x,\ x]$	`[16,18,23,28,30,35]`

Table 7: Wyckoff site: `48g`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,4,42,43]`
2	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[2,3,41,44]`
3	$\left[\frac{1}{4},\ x,\ \frac{1}{4}\right]$	`[5,8,38,39]`
4	$\left[\frac{1}{4},\ -x,\ \frac{3}{4}\right]$	`[6,7,37,40]`
5	$\left[\frac{1}{4},\ \frac{1}{4},\ x\right]$	`[9,12,45,48]`
6	$\left[\frac{3}{4},\ \frac{1}{4},\ -x\right]$	`[10,11,46,47]`
7	$\left[\frac{1}{4},\ x,\ \frac{3}{4}\right]$	`[13,16,30,31]`
8	$\left[\frac{3}{4},\ -x,\ \frac{3}{4}\right]$	`[14,15,29,32]`
9	$\left[x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[17,20,26,27]`
10	$\left[-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[18,19,25,28]`
11	$\left[\frac{1}{4},\ \frac{1}{4},\ -x\right]$	`[21,24,33,36]`
12	$\left[\frac{1}{4},\ \frac{3}{4},\ x\right]$	`[22,23,34,35]`

Table 8: Wyckoff site: `48h`, site symmetry: `m.m2`

No.	position	mapping
1	$[0,\ y,\ y]$	`[1,18,28,43]`
2	$[0,\ -y,\ y]$	`[2,20,27,41]`
3	$[0,\ y,\ -y]$	`[3,17,26,44]`
4	$[0,\ -y,\ -y]$	`[4,19,25,42]`
5	$[y,\ 0,\ y]$	`[5,15,32,38]`
6	$[y,\ 0,\ -y]$	`[6,13,31,40]`
7	$[-y,\ 0,\ y]$	`[7,16,30,37]`
8	$[-y,\ 0,\ -y]$	`[8,14,29,39]`
9	$[y,\ y,\ 0]$	`[9,21,36,48]`
10	$[-y,\ y,\ 0]$	`[10,23,35,46]`
11	$[y,\ -y,\ 0]$	`[11,22,34,47]`
12	$[-y,\ -y,\ 0]$	`[12,24,33,45]`

Table 9: Wyckoff site: `48i`, site symmetry: `m.m2`

No.	position	mapping
1	$\left[\frac{1}{2}, y, y\right]$	`[1,18,28,43]`
2	$\left[\frac{1}{2}, -y, y\right]$	`[2,20,27,41]`
3	$\left[\frac{1}{2}, y, -y\right]$	`[3,17,26,44]`
4	$\left[\frac{1}{2}, -y, -y\right]$	`[4,19,25,42]`
5	$\left[y, \frac{1}{2}, y\right]$	`[5,15,32,38]`
6	$\left[y, \frac{1}{2}, -y\right]$	`[6,13,31,40]`
7	$\left[-y, \frac{1}{2}, y\right]$	`[7,16,30,37]`
8	$\left[-y, \frac{1}{2}, -y\right]$	`[8,14,29,39]`
9	$\left[y, y, \frac{1}{2}\right]$	`[9,21,36,48]`
10	$\left[-y, y, \frac{1}{2}\right]$	`[10,23,35,46]`
11	$\left[y, -y, \frac{1}{2}\right]$	`[11,22,34,47]`
12	$\left[-y, -y, \frac{1}{2}\right]$	`[12,24,33,45]`

Table 10: Wyckoff site: `96j`, site symmetry: `m..`

No.	position	mapping
1	$[0, y, z]$	`[1,28]`
2	$[0, -y, z]$	`[2,27]`
3	$[0, y, -z]$	`[3,26]`
4	$[0, -y, -z]$	`[4,25]`
5	$[z, 0, y]$	`[5,32]`
6	$[z, 0, -y]$	`[6,31]`
7	$[-z, 0, y]$	`[7,30]`
8	$[-z, 0, -y]$	`[8,29]`
9	$[y, z, 0]$	`[9,36]`
10	$[-y, z, 0]$	`[10,35]`
11	$[y, -z, 0]$	`[11,34]`
12	$[-y, -z, 0]$	`[12,33]`
13	$[y, 0, -z]$	`[13,40]`
14	$[-y, 0, -z]$	`[14,39]`
15	$[y, 0, z]$	`[15,38]`
16	$[-y, 0, z]$	`[16,37]`
17	$[0, z, -y]$	`[17,44]`
18	$[0, z, y]$	`[18,43]`
19	$[0, -z, -y]$	`[19,42]`
20	$[0, -z, y]$	`[20,41]`
21	$[z, y, 0]$	`[21,48]`
22	$[z, -y, 0]$	`[22,47]`
23	$[-z, y, 0]$	`[23,46]`
24	$[-z, -y, 0]$	`[24,45]`

Table 11: Wyckoff site: `96k`, site symmetry: `..m`

No.	position	mapping
1	$[x,\ x,\ z]$	`[1,38]`
2	$[-x,\ -x,\ z]$	`[2,37]`
3	$[-x,\ x,\ -z]$	`[3,39]`
4	$[x,\ -x,\ -z]$	`[4,40]`
5	$[z,\ x,\ x]$	`[5,48]`
6	$[z,\ -x,\ -x]$	`[6,47]`
7	$[-z,\ -x,\ x]$	`[7,45]`
8	$[-z,\ x,\ -x]$	`[8,46]`
9	$[x,\ z,\ x]$	`[9,43]`
10	$[-x,\ z,\ -x]$	`[10,44]`
11	$[x,\ -z,\ -x]$	`[11,42]`
12	$[-x,\ -z,\ x]$	`[12,41]`
13	$[x,\ x,\ -z]$	`[13,26]`
14	$[-x,\ -x,\ -z]$	`[14,25]`
15	$[x,\ -x,\ z]$	`[15,27]`
16	$[-x,\ x,\ z]$	`[16,28]`
17	$[x,\ z,\ -x]$	`[17,36]`
18	$[-x,\ z,\ x]$	`[18,35]`
19	$[-x,\ -z,\ -x]$	`[19,33]`
20	$[x,\ -z,\ x]$	`[20,34]`
21	$[z,\ x,\ -x]$	`[21,31]`
22	$[z,\ -x,\ x]$	`[22,32]`
23	$[-z,\ x,\ x]$	`[23,30]`
24	$[-z,\ -x,\ -x]$	`[24,29]`

Table 12: Wyckoff site: `192l`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-x,\ -y,\ z]$	`[2]`
3	$[-x,\ y,\ -z]$	`[3]`
4	$[x,\ -y,\ -z]$	`[4]`
5	$[z,\ x,\ y]$	`[5]`
6	$[z,\ -x,\ -y]$	`[6]`
7	$[-z,\ -x,\ y]$	`[7]`
8	$[-z,\ x,\ -y]$	`[8]`
9	$[y,\ z,\ x]$	`[9]`
10	$[-y,\ z,\ -x]$	`[10]`
11	$[y,\ -z,\ -x]$	`[11]`
12	$[-y,\ -z,\ x]$	`[12]`
13	$[y,\ x,\ -z]$	`[13]`
14	$[-y,\ -x,\ -z]$	`[14]`
15	$[y,\ -x,\ z]$	`[15]`
16	$[-y,\ x,\ z]$	`[16]`
17	$[x,\ z,\ -y]$	`[17]`

continued ...

Table 12

No.	position	mapping
18	$[-x, z, y]$	[18]
19	$[-x, -z, -y]$	[19]
20	$[x, -z, y]$	[20]
21	$[z, y, -x]$	[21]
22	$[z, -y, x]$	[22]
23	$[-z, y, x]$	[23]
24	$[-z, -y, -x]$	[24]
25	$[-x, -y, -z]$	[25]
26	$[x, y, -z]$	[26]
27	$[x, -y, z]$	[27]
28	$[-x, y, z]$	[28]
29	$[-z, -x, -y]$	[29]
30	$[-z, x, y]$	[30]
31	$[z, x, -y]$	[31]
32	$[z, -x, y]$	[32]
33	$[-y, -z, -x]$	[33]
34	$[y, -z, x]$	[34]
35	$[-y, z, x]$	[35]
36	$[y, z, -x]$	[36]
37	$[-y, -x, z]$	[37]
38	$[y, x, z]$	[38]
39	$[-y, x, -z]$	[39]
40	$[y, -x, -z]$	[40]
41	$[-x, -z, y]$	[41]
42	$[x, -z, -y]$	[42]
43	$[x, z, y]$	[43]
44	$[-x, z, -y]$	[44]
45	$[-z, -y, x]$	[45]
46	$[-z, y, -x]$	[46]
47	$[z, -y, -x]$	[47]
48	$[z, y, x]$	[48]